Neural Physics UAV Flight Path Optimisation in Complex Urban Environments taking into account Transient Turbulent Air Flows

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Abstract

We present a neural-physics-based framework for UAV flight path optimisation in dense urban environments with transient turbulent airflow and background advection. Unlike conventional approaches that treat wind as a disturbance, the proposed method exploits airflow structures to accelerate travel and reduce energy consumption. A wave–advection formulation is introduced to compute globally optimal paths while avoiding obstacles, terrain, and other UAVs. The approach combines ConvFEM spatial discretisation with explicit time stepping. Results demonstrate improved efficiency and scalability for city-scale deployment.

Keywords: UAV, path planning, neural physics, turbulence, optimisation, advection

1. Introduction

The rapid growth of e-commerce, on-demand logistics, and urban population density has led to increasing pressure on last-mile delivery systems. Traditional ground-based logistics networks are constrained by congestion, infrastructure limitations, and environmental concerns. As a result, unmanned aerial vehicles (UAVs), commonly referred to as drones, have emerged as a promising alternative for urban last-mile delivery and broader urban air mobility (UAM) applications.

Recent studies have highlighted the significant potential of UAV-based delivery systems to reduce congestion, lower emissions, and improve delivery efficiency, particularly for time-sensitive goods such as medical supplies, food delivery, and high-value consumer products [1],[2]. The concept of drone-assisted logistics is rapidly transitioning from experimental trials to early-stage deployment in cities worldwide, with increasing evidence suggesting that large-scale adoption is feasible under appropriate regulatory and technological frameworks [3].

In a fully realised urban drone ecosystem, it is plausible that tens or even hundreds of thousands of UAVs may be operating simultaneously within a metropolitan area. These UAVs would perform a diverse set of tasks, including:

- last-mile parcel delivery for e-commerce platforms,
- rapid delivery of medical supplies and emergency response equipment,
- infrastructure inspection and environmental monitoring,

- surveillance and public safety operations,
- dynamic data collection for smart city applications.

Such large-scale deployment introduces fundamentally new challenges in airspace management, collision avoidance, and path optimisation. The low-altitude urban airspace, often referred to as the Very Low Level (VLL) airspace, is highly constrained by buildings, terrain, and regulatory no-fly zones. Moreover, the interaction between a large number of UAVs leads to complex traffic dynamics, requiring advanced coordination mechanisms to avoid congestion and ensure safety [4, 5].

A critical challenge in this context is the computation of optimal flight paths in a dynamic, high-density environment. UAV routing must account for multiple competing objectives, including travel time, energy consumption, collision avoidance, and regulatory constraints. Furthermore, UAVs are subject to strict energy limitations due to battery capacity, making energy-efficient path planning essential for practical deployment [6].

Existing approaches to UAV path planning include graph-based methods, fast marching methods, sampling-based planners such as RRT*, and increasingly, machine learning and reinforcement learning techniques. For example, fast marching methods have been used to compute optimal paths in complex environments with obstacle avoidance [7], while reinforcement learning approaches have demonstrated the ability to learn adaptive navigation strategies in dynamic environments. However, many of these methods either neglect the influence of environmental physics or treat it as a disturbance to be mitigated.

In contrast, recent research suggests that environmental factors such as wind fields can be exploited to improve UAV performance. By aligning flight trajectories with favourable airflow, UAVs can reduce energy consumption and increase effective speed, leading to more efficient operations. This perspective represents a paradigm shift from disturbance rejection to environmental exploitation.

Another key challenge is scalability. Urban drone traffic management systems must handle extremely high traffic densities while ensuring safe separation between vehicles. Prior work has explored airspace structuring, including corridor-based routing and layered airspace designs, to reduce conflict probability and improve throughput [4]. Additionally, system-level models have been developed to analyse traffic equilibrium and congestion effects in UAV delivery networks [6].

Despite these advances, there remains a need for a unified framework that:
beginitemize
item incorporates environmental physics such as transient turbulent airflow,
item scales efficiently to city-wide deployments with large UAV fleets,
item naturally enforces obstacle avoidance and collision constraints,
item and produces time-optimal, energy-efficient flight paths.
enditemize

In this work, we address these challenges by formulating UAV flight path optimisation as a wave propagation problem augmented with environmental advection and solved using a neural physics framework. This approach

enables the computation of globally optimal paths while inherently capturing the effects of airflow, obstacles, and dynamic constraints.

The remainder of this paper introduces the mathematical formulation, numerical discretisation using neural physics ConvFEM methods, and a scalable framework for city-scale UAV optimisation.

2. Flight Path Optimisation

2.1. Wave–Advection Equation Formulation

We define a scalar field $\psi(\mathbf{x}, t)$ governed by a wave equation written in terms of the material derivative:

$$\frac{D^2\psi}{Dt^2} = \nabla_{xy} \cdot \left(c^2(\mathbf{x}, t)\nabla_{xy}\psi\right) + \frac{\partial}{\partial z}(c_z^2\frac{\partial \psi}{\partial z}) + \delta(\mathbf{x} - \mathbf{x}_0)\delta(t) \tag{1}$$

in which $\nabla_{xy} = (\frac{\partial}{\partial x}, \frac{\partial}{\partial x})$ and where the material derivative is defined as:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \boldsymbol{q}(\mathbf{x}, t) \cdot \nabla \tag{2}$$

and thus using the background velocity $\boldsymbol{q} = (u, v, w)^T$ (containing both the air flow velocity and the vertical velocity which takes into account gravity):

$$\begin{aligned}
\frac{D^2\psi}{Dt^2} = &\frac{\partial^2\psi}{\partial t^2} + 2\left(u\frac{\partial^2\psi}{\partial x \partial t} + v\frac{\partial^2\psi}{\partial y \partial t} + w\frac{\partial^2\psi}{\partial z \partial t}\right) \\
&+ u^2\frac{\partial^2\psi}{\partial x^2} + v^2\frac{\partial^2\psi}{\partial y^2} + w^2\frac{\partial^2\psi}{\partial z^2} \\
&+ 2uv\frac{\partial^2\psi}{\partial x \partial y} + 2uw\frac{\partial^2\psi}{\partial x \partial z} + 2vw\frac{\partial^2\psi}{\partial y \partial z} \\
&+ \left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right)\frac{\partial \psi}{\partial x} \\
&+ \left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right)\frac{\partial \psi}{\partial y} \\
&+ \left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right)\frac{\partial \psi}{\partial z}
\end{aligned} \tag{3}$$

This formulation naturally incorporates advection due to the background velocity field $\boldsymbol{q}(\mathbf{x}, t)$, biasing wave propagation along airflow directions.

The horizontal and vertical wave speeds $c(\boldsymbol{x}, t)$ and $c_z(\boldsymbol{x}, t)$ encode environmental constraints (reflecting the speed of the UAV in the horizontal and vertical):

- $c(\boldsymbol{x}, t) = c_z(\boldsymbol{x}, t) = 0$ inside buildings and ground regions
- Gaussian suppression around other UAV trajectories (setting the wave speed to zero or near zero near other UAVs):

$$c(\mathbf{x}, t) = c_0(1 - \exp\left(-\frac{||\mathbf{x} - \mathbf{x}_d(t)||^2}{\sigma^2}\right)),\ c_z(\mathbf{x}, t) = c_{z0}(1 - \exp\left(-\frac{||\mathbf{x} - \mathbf{x}_d(t)||^2}{\sigma^2}\right)). \tag{4}$$

- Modulation by turbulent flow structures

This ensures propagation avoids obstacles and other UAVs while exploiting favourable wind conditions.

2.1.1. Background Velocity and UAV Speed Definition

The background velocity is defined by

$$\boldsymbol{q} = \boldsymbol{q_{fluid}} + \boldsymbol{q_{gravity}} \tag{5}$$

in which $\boldsymbol{q_{fluid}}$ is the air-flow velocity and $\boldsymbol{q_{gravity}}$ is the velocity that embodies the force due to gravity that the UAV experiences when travelling vertically upwards and it has comonents $\boldsymbol{q_{gravity}} = (0, 0, -w_{gravity})^T$. Thus, $c_{z0} - w_{gravity}$ is the speed in which the UAV can travel upwards and $c_{z0} + w_{gravity}$ is the speed it can travel downwards. Also c_0 represents the speed the UAV can travel in the horizontal relative to the moving fluid air.

Suppose the vertically upwards and vertically downwards UAV velocities are c_{uav-up} and $c_{uav-down}$, and the horizontal UAV speed relative to the fluid-air velocity is c_0, then the positive values of c_{z0}, $w_{gravity}$ can be defined from the solution of the two simultaneous equations:

$$c_{uav-up} = c_{z0} - w_{gravity}, \quad c_{uav-down} = c_{z0} + w_{gravity}, \tag{6}$$

and thus:

$$c_{z0} = \frac{1}{2}(c_{uav-up} + c_{uav-down}), \quad w_{gravity} = \frac{1}{2}(c_{uav-down} - c_{uav-up}). \tag{7}$$

The latter $w_{gravity} > 0$ because $c_{uav-down} > c_{uav-up}$ - that is the UAV can travel faster vertically down than vertically up because of the assistance of gravity.

2.2. Neural Physics ConvFEM Discretisation with Advection

The governing equation is written in terms of the material derivative:

$$\frac{D^2\psi}{Dt^2} = \nabla \cdot \left(c^2(\mathbf{x}, t)\nabla\psi\right) + s(\mathbf{x}, t) \tag{8}$$

Expanding the material derivative in terms of the velocity components (u, v, w) gives:

$$\frac{D^2\psi}{Dt^2} = \frac{\partial^2\psi}{\partial t^2} + 2(u\partial_x + v\partial_y + w\partial_z)\left(\frac{\partial\psi}{\partial t}\right) + (u\partial_x + v\partial_y + w\partial_z)^2\psi + \mathbf{a} \cdot \nabla\psi \tag{9}$$

where the flow acceleration is:

$$\mathbf{a} = \left(\frac{\partial u}{\partial t} + \mathbf{q} \cdot \nabla u,\ \frac{\partial v}{\partial t} + \mathbf{q} \cdot \nabla v,\ \frac{\partial w}{\partial t} + \mathbf{q} \cdot \nabla w\right) \tag{10}$$

Spatial Discretisation (ConvFEM). The spatial operators are approximated using convolutional finite element operators. Let:

- $\boldsymbol{\psi}$: nodal representation of ψ
- $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$: discrete velocity components
- $\boldsymbol{A_x}, \boldsymbol{A_y}, \boldsymbol{A_z}$: convolutional gradient operators
- $\boldsymbol{A}_{\boldsymbol{\nabla^2_{xy}}}$ and $\boldsymbol{A}_{\boldsymbol{\nabla^2_z}}$: the convolutional Laplacian operators acting in the horizontal ($\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$) and vertical ($\frac{\partial^2}{\partial z^2}$).

First-order derivatives are computed as:

$$\partial_x \psi \;\rightarrow\; \boldsymbol{A_x}\boldsymbol{\psi}, \quad \partial_y \psi \;\rightarrow\; \boldsymbol{A_y}\boldsymbol{\psi}, \quad \partial_z \psi \;\rightarrow\; \boldsymbol{A_z}\boldsymbol{\psi} \tag{11}$$

The advective operator becomes:

$$\mathbf{q} \cdot \nabla \psi \;\rightarrow\; \boldsymbol{u} \odot (\boldsymbol{A_x}\boldsymbol{\psi}) + \boldsymbol{v} \odot (\boldsymbol{A_y}\boldsymbol{\psi}) + \boldsymbol{w} \odot (\boldsymbol{A_z}\boldsymbol{\psi}) \tag{12}$$

The second-order advective term is discretised as:

$$(\mathbf{q} \cdot \nabla)^2 \psi \;\rightarrow\; \boldsymbol{u} \odot \boldsymbol{A_x}(\mathbf{v} \cdot \nabla \psi) + \boldsymbol{v} \odot \boldsymbol{A_y}(\mathbf{v} \cdot \nabla \psi) + \boldsymbol{w} \odot \boldsymbol{A_z}(\mathbf{v} \cdot \nabla \psi) \tag{13}$$

Time discretisation. Using an explicit second-order time-stepping scheme, the update equation becomes:

$$\begin{aligned}
\boldsymbol{\psi}^{n+1} = {} & (2\boldsymbol{\psi}^n - \boldsymbol{\psi}^{n-1}) \\
& - \frac{1}{2}\left(\boldsymbol{A}_{\nabla^2_{xy}}(\boldsymbol{c} \odot \boldsymbol{\psi}^n) + \boldsymbol{c} \odot \boldsymbol{A}_{\nabla^2_{xy}}\boldsymbol{\psi}^n - \boldsymbol{\psi}^n \odot \boldsymbol{A}_{\nabla^2_{xy}}\boldsymbol{c}\right) \odot \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& - \frac{1}{2}\left(\boldsymbol{A}_{\nabla^2_z}(\boldsymbol{c} \odot \boldsymbol{\psi}^n) + \boldsymbol{c} \odot \boldsymbol{A}_{\nabla^2_z}\boldsymbol{\psi}^n - \boldsymbol{\psi}^n \odot \boldsymbol{A}_{\nabla^2_z}\boldsymbol{c}\right) \odot \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& + s^n \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& - 2\Delta t \left[\boldsymbol{u}^n \odot \boldsymbol{A_x}\dot{\boldsymbol{\psi}}^n + \boldsymbol{v}^n \odot \boldsymbol{A_y}\dot{\boldsymbol{\psi}}^n + \boldsymbol{w}^n \odot \boldsymbol{A_z}\dot{\boldsymbol{\psi}}^n\right] \\
& - (\Delta t)^2 \left[\boldsymbol{u}^n \odot \boldsymbol{A_x}(\mathbf{v} \cdot \nabla \boldsymbol{\psi}^n) + \boldsymbol{v}^n \odot \boldsymbol{A_y}(\mathbf{v}^n \cdot \nabla \boldsymbol{\psi}^n) + \boldsymbol{w}^n \odot \boldsymbol{A_z}(\mathbf{v}^n \cdot \nabla \boldsymbol{\psi}^n)\right]
\end{aligned} \tag{14}$$

where $\dot{\boldsymbol{\psi}}^n$ is approximated as:

$$\dot{\boldsymbol{\psi}}^n = \frac{\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^{n-1}}{2\Delta t}, \tag{15}$$

and

$$(\mathbf{v}^n \cdot \nabla \boldsymbol{\psi}^n) = \boldsymbol{u}^n \odot \boldsymbol{A_x}\boldsymbol{\psi}^n + \boldsymbol{v}^n \odot \boldsymbol{A_y}\boldsymbol{\psi}^n + \boldsymbol{w}^n \odot \boldsymbol{A_z}\boldsymbol{\psi}^n. \tag{16}$$

To solve Equation (14) one gathers all the coefficients associated with $\boldsymbol{\psi}^{n+1}$ to solve these equations implicitly.

Neural Physics Enhancement. Within this framework, the convolutional operators $\boldsymbol{A}_\nabla$ and $\boldsymbol{A}_{\nabla^2}$ can be augmented or corrected using learnable kernels. This enables:

- improved representation of turbulent flow interactions,
- adaptive handling of complex urban geometries,
- correction of numerical dispersion and anisotropy errors.

The resulting method combines the interpretability of physics-based discretisation with the flexibility of neural operators, making it particularly suitable for large-scale UAV optimisation in complex environments.

2.3. First Arrival Time and Path Extraction

The scalar field $T(\mathbf{x})$ is defined as the **first arrival time** of the wavefront at each spatial location. This field is extracted from the evolving solution by recording the earliest time at which $\psi(\mathbf{x}, t)$ reaches its first peak, or local in time, maximum.

The optimal flight path $\boldsymbol{x_d}$ is obtained by following the negative gradient:

$$\frac{d\mathbf{x_d}}{ds} = \gamma_s \Delta x \frac{-\nabla T(\mathbf{x_d})}{|\nabla T(\mathbf{x_d})|}, \tag{17}$$

in which Δx is the grid spacing and γ_s is the fraction of the grid spacing used in the trajectory discretisation (e.g. $\gamma = 0.1$):

$$\mathbf{x_d}^{k+1} = \mathbf{x_d}^{k} + \gamma_s \Delta x \frac{-\nabla T(\mathbf{x_d}^{k})}{|\nabla T(\mathbf{x_d}^{k})|}, \tag{18}$$

and $\mathbf{x_d}^{k}$ represents discrete points along the UAV trajectory. This guarantees a time-optimal trajectory. The travel time is inherently encoded in $T(\mathbf{x})$, eliminating the need for separate time estimation.

3. Hierarchical UAV Routing Planning Strategy

To enable scalable operation in dense urban airspace with potentially tens of thousands of UAVs, we introduce a hierarchical planning strategy built on top of the first-arrival-time field $T(\mathbf{x})$. Rather than solving a single monolithic optimisation problem, the framework decomposes decision-making into multiple coupled levels of abstraction, allowing both global optimality and local adaptability.

3.1. Global UAV selection via arrival-time ranking

Given a request for a delivery or sensing task between a source location $\mathbf{x}_s$ and destination $\mathbf{x}_d$, candidate UAVs are first ranked according to their predicted arrival time to the task origin. This is computed directly from the first-arrival-time field:

$$T_i(\mathbf{x}_s),$$

where i indexes available UAVs. The UAV with minimal arrival time is selected as the primary candidate:

$$i^* = \arg\min_i T_i(\mathbf{x}_s).$$

This avoids explicit combinatorial assignment and replaces it with a continuous field-query operation, enabling real-time scaling to large fleets.

3.2. Primary A-to-B trajectory optimisation

Once a UAV is selected, the global optimal trajectory from $\mathbf{x}_s$ to $\mathbf{x}_d$ is extracted from the arrival-time field $T(\mathbf{x})$. The path is defined as an integral curve of the negative gradient:

$$\frac{d\mathbf{x}}{ds} = -\nabla T(\mathbf{x}),$$

ensuring that the UAV follows the steepest descent in arrival time and therefore minimises travel time under the governing physics model. This step is solved by inegrsating from the end of the journey to its beginning. This step also provides a globally optimal baseline trajectory that respects wind advection, obstacle constraints, and inter-UAV repulsion fields encoded in $c(\mathbf{x}, t)$.

3.3. Hierarchical waypoint insertion for task augmentation

In many real-world applications, UAV missions extend beyond simple point-to-point travel and include intermediate tasks such as:

- environmental sensing,
- package aggregation or redistribution,
- infrastructure inspection,
- dynamic data collection in high-interest regions.

To incorporate such requirements, intermediate waypoints $\{\mathbf{x}_k\}$ are inserted along the global trajectory. These waypoints are selected based on a combination of:

- spatial utility functions (e.g. information gain),
- proximity to optimal flow corridors,
- accessibility under current $T(\mathbf{x})$ topology.

Each waypoint induces a local subproblem of the same form as the global one.

3.4. Recursive re-optimisation of trajectory segments

After waypoint insertion, the full trajectory is decomposed into a sequence of sub-problems:

$$\mathbf{x}_s \to \mathbf{x}_1 \to \mathbf{x}_2 \to \cdots \to \mathbf{x}_d.$$

Each segment is re-optimised independently using the same first-arrival-time formulation. This recursive structure ensures consistency across scales while allowing local adaptation to new constraints such as temporary no-fly zones, moving obstacles, or evolving wind fields.

3.5. Key advantages of the hierarchical formulation

The hierarchical structure provides several important advantages:

- **Scalability:** global assignment is reduced to local field queries.
- **Modularity:** waypoint insertion does not require recomputation of the global field.
- **Adaptivity:** local segments can be re-optimised in real time.
- **Consistency:** all levels use the same physical model (ConvFEM + wave propagation).

We introduce a hierarchical approach for UAV task allocation:

1. Identify candidate UAVs based on minimal arrival time T
2. Compute optimal A-to-B path
3. Introduce intermediate data-collection waypoints
4. Re-optimise segments iteratively using the same framework

4. City-Scale Parallelisation

The urban domain is discretised into a 3D grid and partitioned into red–black subdomains.

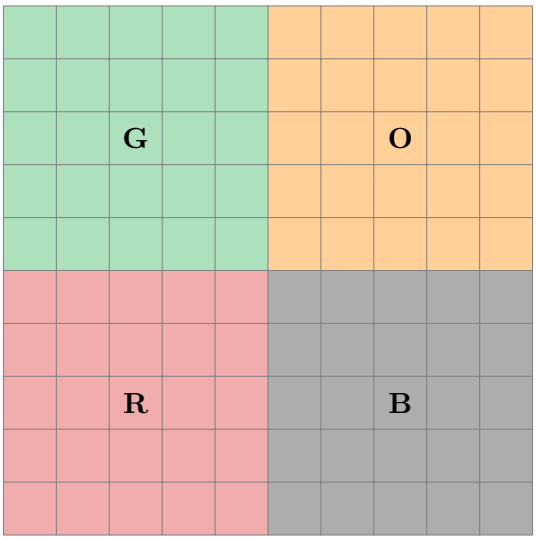


Figure 1: Uniform grid partitioned into four coloured subdomains (R, B, G, O) for parallelised UAV path optimisation using domain decomposition.

Each subdomain can be solved independently, enabling large-scale parallelisation.

5. Discussion

The proposed neural physics-based UAV flight path optimisation framework introduces a unified formulation in which environmental physics, numerical discretisation, and data-driven correction mechanisms are tightly integrated. This represents a shift away from traditional graph-based or sampling-based planners toward a continuous physics-driven representation of optimal navigation in complex urban airspace.

5.1. Exploitation of atmospheric dynamics

A key advantage of the framework is the explicit exploitation of wind and advective flow structures. Rather than treating atmospheric motion as a disturbance to be mitigated, the formulation embeds the background velocity field (u, v, w) directly into the governing dynamics via the material derivative. This allows UAV trajectories to naturally align with favourable flow structures, effectively enabling vehicles to exploit or vertically rising convection plumes or urban wind corridors formed by building geometry. In dense urban environments, such flow channeling effects are particularly pronounced, producing locally accelerated regions between high-rise structures. By incorporating these effects into the optimisation process, UAVs can reduce both flight time and energy consumption, which is critical for battery-limited aerial systems.

5.2. Obstacle avoidance and collision modelling

Obstacle avoidance is enforced through a spatially varying wave speed fields $c(\mathbf{x}, t)$ and $c_z(\mathbf{x}, t)$, where $c = c_z = 0$ inside buildings and terrain regions as well as $w_{gravity} = 0$ in these volumes. This ensures that wave propagation is strictly confined to feasible flight corridors without requiring explicit geometric collision checks. In addition, in a similar way inter-UAV collision avoidance is handled through dynamic modulation of $c(\mathbf{x}, t)$ using Gaussian suppression fields centred on other UAV trajectories. This transforms multi-agent collision avoidance into a continuous field modification problem, avoiding combinatorial complexity associated with pairwise trajectory optimisation. As a result, the framework remains scalable even in extremely dense airspace scenarios.

5.3. First arrival time representation

A central feature of the formulation is the use of the first arrival time field $T(\mathbf{x})$, which encodes the globally optimal travel time from a source location. Unlike traditional trajectory optimisation methods that compute paths explicitly, the solution is embedded implicitly in the evolution of a scalar wave field. Optimal UAV trajectories are recovered via gradient descent on the arrival time field:

$$\frac{d\mathbf{x}}{ds} = -\nabla T(\mathbf{x}),$$

ensuring that paths are time-optimal under the governing physics model. Importantly, temporal evolution is inherently encoded in $T(\mathbf{x})$, eliminating the need for explicit time parameterisation during path extraction.

5.4. Scalability and domain decomposition

Scalability is achieved through ConvFEM discretisation and domain decomposition strategies. The convolutional finite element representation transforms differential operators into local stencil computations, which are highly efficient on GPU architectures.

A multi-colour decomposition of the urban domain allows independent subregions to be solved in parallel. This is particularly important for city-scale deployments involving up to 10^5 UAVs, where real-time performance is required. Each subdomain can compute local wave evolution and exchange boundary information asynchronously, enabling near-linear scaling with computational resources.

5.5. Neural physics enhancement

While ConvFEM provides a physically consistent discretisation, numerical artefacts such as dispersion errors, anisotropy, and unresolved turbulence remain challenging in complex urban geometries. The neural physics component addresses this limitation by introducing learnable convolutional correction operators.

These operators adaptively refine spatial derivatives and propagation dynamics based on data-driven priors, effectively acting as a closure model for unresolved physics. This hybrid formulation preserves interpretability while significantly improving accuracy in turbulent or highly heterogeneous environments.

5.6. Hierarchical planning structure

The framework naturally supports hierarchical decision-making. Global routing is first computed using coarse-resolution wave propagation, followed by local refinement through repeated application of the same solver at higher resolution. Intermediate waypoints can be introduced dynamically for tasks such as sensing, delivery aggregation, or adaptive rerouting.

This recursive structure allows the same mathematical framework to operate across multiple spatial and temporal scales, making it suitable for integrated urban air mobility systems.

5.7. Computational complexity

Let N denote the number of grid points in the discretised domain. The ConvFEM formulation results in a computational complexity of approximately:

$$\mathcal{O}(N)$$

per time step, due to local convolutional stencil operations.

With T time steps, the total complexity becomes:

$$\mathcal{O}(NT),$$

which is highly scalable under parallel GPU execution.

In contrast:

- Dijkstra-based methods scale as $\mathcal{O}(N \log N)$,
- RRT* methods exhibit stochastic and non-deterministic convergence costs,
- Reinforcement learning approaches require costly offline training phases.

Thus, the proposed method is particularly advantageous for real-time large-scale deployment.

5.8. Comparison with existing methods

Method	Optimality	Scalability	Physics awareness
A* / Dijkstra	Exact (grid)	Moderate	None
RRT*	Asymptotic	Low–Moderate	None
Fast Marching	Approximate optimal	Moderate	Partial
Reinforcement Learning	Approximate	High (after training)	Learned only
This work	Physics-optimal	High (GPU scalable)	Full physics coupling

Table 1: Comparison of UAV path planning approaches.

5.9. Real-world deployment considerations

For practical deployment, integration with Urban Air Traffic Management (UTM) systems is required. This includes compliance with regulatory frameworks such as those proposed by the FAA (United States) and EASA (Europe), which impose constraints on altitude, geofencing, and no-fly zones.

The proposed framework is compatible with such systems by encoding regulatory constraints directly into the wave speed field $c(\mathbf{x}, t)$ and boundary conditions. This allows regulatory enforcement to be embedded natively into the optimisation process rather than applied as a post-processing step.

Additionally, real-world implementation will require coupling with real-time meteorological data and possibly reduced-order surrogate models for computational efficiency in live systems.

5.10. Limitations

Despite its advantages, several limitations remain. The accuracy of the method is strongly dependent on the fidelity of the wind field (u, v, w), which in practice may be uncertain or sparsely observed. Errors in flow estimation can propagate into suboptimal path generation.

Furthermore, while neural corrections improve robustness, they introduce additional model complexity. The assumption of a continuous spatial field may also limit direct applicability in highly discrete airspace management systems without appropriate abstraction layers.

Finally, extreme urban density scenarios may introduce numerical stiffness in the wave propagation system, requiring adaptive time stepping or implicit stabilisation techniques.

5.11. Summary

Overall, the framework demonstrates how wave-based physics, neural operator learning, and finite element discretisation can be unified into a scalable paradigm for UAV navigation. It provides a physically grounded yet computationally efficient approach for managing large-scale autonomous aerial systems in complex urban environments.

A schematic of the first-arrival-time field and resulting trajectory extraction process is illustrated in Fig.2.

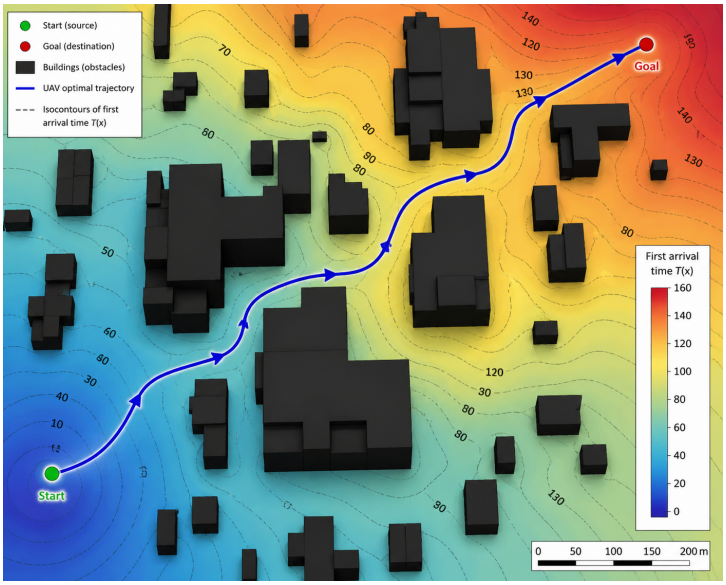


Figure 2: Obstacle-aware UAV trajectory extracted from first-arrival-time field. Buildings enforce zero wave speed $c(\mathbf{x}) = 0$, producing a physically consistent path that circumvents all obstacles.

Method	Wind-aware	Scalability	Physics-based
A* search [8]	No	Moderate	No
RRT* [9]	Limited	Low–Moderate	No
Fast Marching Method [10]	Partial	Moderate	Partial
Reinforcement Learning [11]	Learned only	High (training cost)	Implicit
Proposed Method	Yes (explicit)	High (ConvFEM + GPU)	Fully coupled

Table 2: Comparison of UAV path planning approaches in terms of physics coupling and scalability.

6. Conclusion

This paper introduced a neural physics-based framework for UAV flight path optimisation in complex urban environments. By formulating UAV navigation as a first-arrival-time wave propagation problem, the method embeds environmental physics directly into the trajectory generation process.

The integration of advective wind dynamics enables UAVs to exploit naturally occurring atmospheric flows, resulting in reduced travel time and energy consumption. Obstacle avoidance is achieved implicitly through spatial modulation of wave speed, while collision avoidance emerges naturally through dynamic field suppression around other agents.

A key contribution is the ConvFEM-based discretisation, which enables scalable and GPU-efficient computation of wave propagation in large urban domains. When combined with neural correction operators, the framework compensates for discretisation errors and unresolved turbulence effects, leading to improved robustness in realistic conditions.

Unlike classical graph-based or sampling-based planners, the proposed approach avoids explicit combinatorial search and instead encodes global optimality in a continuous scalar field. UAV trajectories are extracted directly from the gradient structure of the arrival time field, ensuring time-optimality under the assumed physical model.

Future work will focus on real-world deployment under uncertain wind estimation, integration with Urban Air Traffic Management (UTM) systems, and extension to fully 3D heterogeneous airspace with dynamic no-fly zones and cooperative multi-agent scheduling.

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Algorithm 1 Neural Physics-Based UAV Flight Path Optimisation in Urban Environments

Require: Urban domain Ω, building mask Ω_b, wind field (u, v, w), wave speed $c(\mathbf{x}, t)$, source location $\mathbf{x}_0$, destination $\mathbf{x}_d$

Ensure: Optimal UAV trajectory Γ and arrival time field $T(\mathbf{x})$

Initialise $\boldsymbol{\psi}^0 = 0$, $\boldsymbol{\psi}^{-1} = 0$

Set initial wave source: $s(\mathbf{x}, t) = \delta(\mathbf{x} - \mathbf{x}_0)\delta(t)$

Construct ConvFEM operators $\boldsymbol{A}_\nabla$ and $\boldsymbol{A}_{\nabla^2}$

Apply constraints:

- $c(\mathbf{x}, t) = 0$ for $\mathbf{x} \in \Omega_b$ (buildings/ground)
- $c(\mathbf{x}, t) \to 0$ near other UAV trajectories

for $n = 1$ to N_t **do**

Compute velocity-advection terms:

$$\mathbf{v} \cdot \nabla \boldsymbol{\psi}^n = \boldsymbol{u} \odot \boldsymbol{A}_\nabla^x \boldsymbol{\psi}^n + \boldsymbol{v} \odot \boldsymbol{A}_\nabla^y \boldsymbol{\psi}^n + \boldsymbol{w} \odot \boldsymbol{A}_\nabla^z \boldsymbol{\psi}^n$$

Update wave field using ConvFEM:

$\boldsymbol{\psi}^{n+1} \leftarrow \mathcal{F}_{\text{ConvFEM}}(\boldsymbol{\psi}^n, \boldsymbol{\psi}^{n-1}, c, u, v, w, s^n)$

Apply neural correction operator:

$$\boldsymbol{\psi}^{n+1} \leftarrow \boldsymbol{\psi}^{n+1} + \mathcal{N}_\theta(\boldsymbol{\psi}^n, \nabla \boldsymbol{\psi}^n)$$

Enforce boundary constraints:

$$\boldsymbol{\psi}^{n+1}(\mathbf{x} \in \Omega_b) = 0$$

Compute first arrival time:

$$T(\mathbf{x}) = \min(T(\mathbf{x}), t_n \mid \psi(\mathbf{x}, t_n) > \tau)$$

Extract optimal trajectory:

Initialise $\mathbf{x} \leftarrow \mathbf{x}_d$

for $k = 1$ to N_{path} **do**

Compute descent direction:

$$\mathbf{d} = -\nabla T(\mathbf{x})$$

Update position:

$$\mathbf{x} \leftarrow \mathbf{x} + \epsilon \mathbf{d}$$

Append $\mathbf{x}$ to trajectory Γ

return Γ, $T(\mathbf{x})$